

AISSAOUI Ismail

AI & Data Science Engineer

Email: ismail.aissaoui.pro@gmail.com Phone: +213 660 70 77 96

Portfolio: ismail-aissaoui.vercel.app Location: Chlef, Algeria

LinkedIn: linkedin.com/in/aissaoui-ismail-6a77a92a7 GitHub: github.com/i-aissaoui

PROFILE

AI & Data Science Engineer with a background in Full-Stack Software Engineering. Experienced in building projects across diverse domains, including Industrial Digital Twins, Healthcare, Financial Trading, Computer Vision, and Enterprise Management Systems. My primary interests include Deep Learning, Graph Neural Networks, Reinforcement Learning, and Natural Language Processing (NLP).

SKILLS

Programming: Python, TypeScript, JavaScript, Java, SQL

ML / DL: PyTorch, TensorFlow, Keras, Scikit-learn, Lightning, ONNX

AI Techniques: Transformers (BERT, GPT, RAG), GNN, GAN, XGBoost, LSTM

Optimization: PSO, Genetic Algorithm, ACO, Simulated Annealing, Bayesian Optimization

Web & API: React, Next.js, FastAPI, Node.js, TailwindCSS

Data & MLOps: MLflow, Airflow, Kubeflow, DVC, Pandas, NumPy

Databases: PostgreSQL, MongoDB, MySQL, Neo4j, Cassandra

Cloud & DevOps: Docker, Linux, Git, CI/CD, CUDA, AWS, Grafana

PROFESSIONAL EXPERIENCE

Graduate AI Research Engineer (Intern)

Jan 2026 – Jul 2026

Sonatrach (Industrial Surveillance Division) — Algeria

Architected an AI-native Digital Twin for real-time monitoring of Gas Turbines.

- Architected a 3-tier Dual-Track framework decoupling safety-critical Edge models from Cloud analytics.
- Developed **TCNEdge** achieving **0.049ms INT8 ONNX C++ inference** for emergency shutdown.
- Engineered **MambaTS** with a missingness-aware gate to maintain robustness during sensor dropouts.
- Implemented **KAN-PCNN** and **ST-KDFormer** to ensure strict thermodynamic consistency at scale.

Network Engineering Intern

Sep 2024

Algérie Télécom RMC Center — Chlef, Algeria

EDUCATION

National Higher School of Computer Science (ESI-SBA)

2021–2026

Engineering and Master's Degree in Computer Science

Specialization: Artificial Intelligence & Data Science

CERTIFICATIONS

DeepLearning.AI: Machine Learning Specialization, Deep Learning Specialization, NLP Specialization

ZTM Academy: PyTorch: Zero to Mastery, TensorFlow Developer

ACADEMIC THESES

State Engineering Degree Thesis:

Physics-Informed Explainable Digital Twin for Industrial Gas Turbines: A Dual-Track Cloud-Edge Deep Learning Framework with Koopman-Augmented State-Space Models

Master's Degree Thesis:

Hemodynamic Digital Twins: Fusing Hemodynamic Physics-Informed Neural Networks and Mamba-SSM for ICU Patient Prognostics

PROJECTS

Industrial Gas Turbine Digital Twin (Sonatrach)

2026

- **Edge & Cloud Architecture:** Engineered a 3-tier framework decoupling safety-critical TCNEdge (Siemens PLC) from near-edge MambaTS and cloud-based ST-KDFormer.
- **Ultra-Low Latency Inference:** Compiled 1D Dilated Convolutions (TCNEdge) to INT8 ONNX C++, achieving 0.049ms latency for emergency shutdown.
- **Physcis & Robustness:** Implemented MambaTS with missingness-aware gating for broken sensors, and KAN-PCNN with PyTorch Autograd for absolute thermodynamic consistency.

Tech: MambaTS, TCNEdge, ST-KDFormer, KAN-PCNN, PyTorch, C++ ONNX, PINN

Cardiovascular Medical Digital Twin

2026

- **Patient-Level Forecasting:** Predicted Mean Blood Pressure (MBP) stability using 100Hz VitalDB ICU telemetry, implementing 70/15/15 splitting to eliminate data leakage.
- **Physics-Informed AI:** Designed a 0D Windkessel **PI_DeepONet** (Branch/Trunk Nets) penalizing violations of arterial compliance and vascular resistance.
- **Hybrid Edge-Cloud Models:** Architected linear-time **Medical_MambaTS** for bedside monitors and **ST-KDFormer** for deep spatio-temporal attention mapping in the cloud.

Tech: PyTorch, Medical_MambaTS, PI_DeepONet, ST-KDFormer, Hemodynamics, VitalDB

Aura — Enterprise AI Trading Platform

2025

- **Hybrid Forecasting:** Built an autonomous trading system combining LSTM networks for time-series predictions.
- **Strategic Decision-Making:** Implemented Deep Q-Learning (Reinforcement Learning) to optimize long-term reward policies.
- **Real-Time Dashboard:** Engineered a high-throughput FastAPI backend with WebSockets and a Next.js interactive UI.

Tech: TensorFlow, PyTorch, FastAPI, Next.js, WebSockets, RL

GNN-based Recruiting System (Career Connect)

2024–2025

- **Relational Learning:** Developed a recommendation engine using Graph Neural Networks (GNN) to map candidate-job relationships.
- **Semantic Embeddings:** Utilized PyTorch Geometric and SBERT for deep semantic text processing.
- **System Architecture:** Deployed via FastAPI and PostgreSQL, significantly outperforming keyword-based heuristics.

Tech: PyTorch Geometric, SBERT, FastAPI, PostgreSQL, GNN

LANGUAGES

Arabic: Native | English: Advanced | French: Professional